

Gasiorowicz. Bölüm 21

Zaman Bağımlı Pert.

$$H\psi = E\psi \quad H = (H_0 + \lambda H_1)\psi$$

$$\text{Schr} \Rightarrow i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = E\psi \Rightarrow$$

$$i\hbar \frac{\partial \psi}{\partial t} = E\psi \Rightarrow \frac{\partial \psi}{\psi} = \frac{E dt}{i\hbar} \Rightarrow \ln \psi = \frac{E t}{i\hbar} + \ln C \Rightarrow \frac{\ln \psi}{C} = \frac{E t}{i\hbar} \Rightarrow \frac{\ln \psi}{C} = e^{-iEt/\hbar}$$

$$\Rightarrow \frac{\psi}{C} = e^{-iEt/\hbar}$$

$$\Rightarrow \psi = C e^{-iEt/\hbar}$$

$$|\psi\rangle = C e^{-iEt/\hbar} |\phi_n\rangle$$

$$|\psi\rangle = \sum_n C_n(t) e^{-iE_n^0 t/\hbar} |\phi_n\rangle, \quad \langle \phi_n | H | \phi_n \rangle = \langle H \rangle = ?$$

$$H = H_0 + \lambda H_1 \Rightarrow H_1 = V(t)$$

$$i\hbar \frac{\partial \psi}{\partial t} = E\psi \quad i\hbar \left\{ -i \frac{E_n^0}{\hbar} \sum_n C_n(t) e^{-iE_n^0 t/\hbar} |\phi_n\rangle + \sum_n \dot{C}_n(t) e^{-iE_n^0 t/\hbar} |\phi_n\rangle \right\} =$$

$$= (H_0 + \lambda H_1) \left(\sum_n C_n(t) e^{-iE_n^0 t/\hbar} |\phi_n\rangle \right)$$

$$\Rightarrow E_n^0 \sum_n C_n(t) e^{-iE_n^0 t/\hbar} |\phi_n\rangle + i\hbar \sum_n \dot{C}_n(t) e^{-iE_n^0 t/\hbar} |\phi_n\rangle = \sum_n C_n(t) e^{-iE_n^0 t/\hbar} H_0 |\phi_n\rangle + \lambda \sum_n C_n(t) e^{-iE_n^0 t/\hbar} H_1 |\phi_n\rangle$$

$\uparrow V(t)$

$$\langle \phi_m | \Rightarrow$$

$$E_n^0 \sum_n C_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | \phi_n \rangle + i\hbar \sum_n \dot{C}_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | \phi_n \rangle = \sum_n C_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | H_0 | \phi_n \rangle + \lambda \sum_n C_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | V | \phi_n \rangle$$

$\underbrace{\langle \phi_m | \phi_n \rangle}_{\delta_{mn}}$

$$\Rightarrow \sum_n \dot{C}_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | \phi_n \rangle = \frac{\lambda}{i\hbar} \sum_n C_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | V | \phi_n \rangle$$

$$\sum_n \frac{dC_n(t)}{dt} e^{-iE_n^0 t/\hbar} \langle \phi_m | \phi_n \rangle = \frac{\lambda}{i\hbar} \sum_n C_n(t) e^{-iE_n^0 t/\hbar} \langle \phi_m | V | \phi_n \rangle$$

$$\lambda^0 C_n^0(t) + \lambda^1 C_n^1(t) + \lambda^2 C_n^2(t)$$

$$\left[\lambda^0 C_n^0(t) + \lambda^1 C_n^1(t) + \lambda^2 C_n^2(t) \right] e^{-iE_n^0 t/\hbar} = \frac{\lambda}{i\hbar} \sum_n \left[\lambda^0 C_n^0(t) + \lambda^1 C_n^1(t) + \lambda^2 C_n^2(t) \right] e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V | \phi_n \rangle$$

$$\lambda^0 \Rightarrow C_n^0(t) = 0 = \text{start} = C_n^0(t) = \delta_{mn}$$

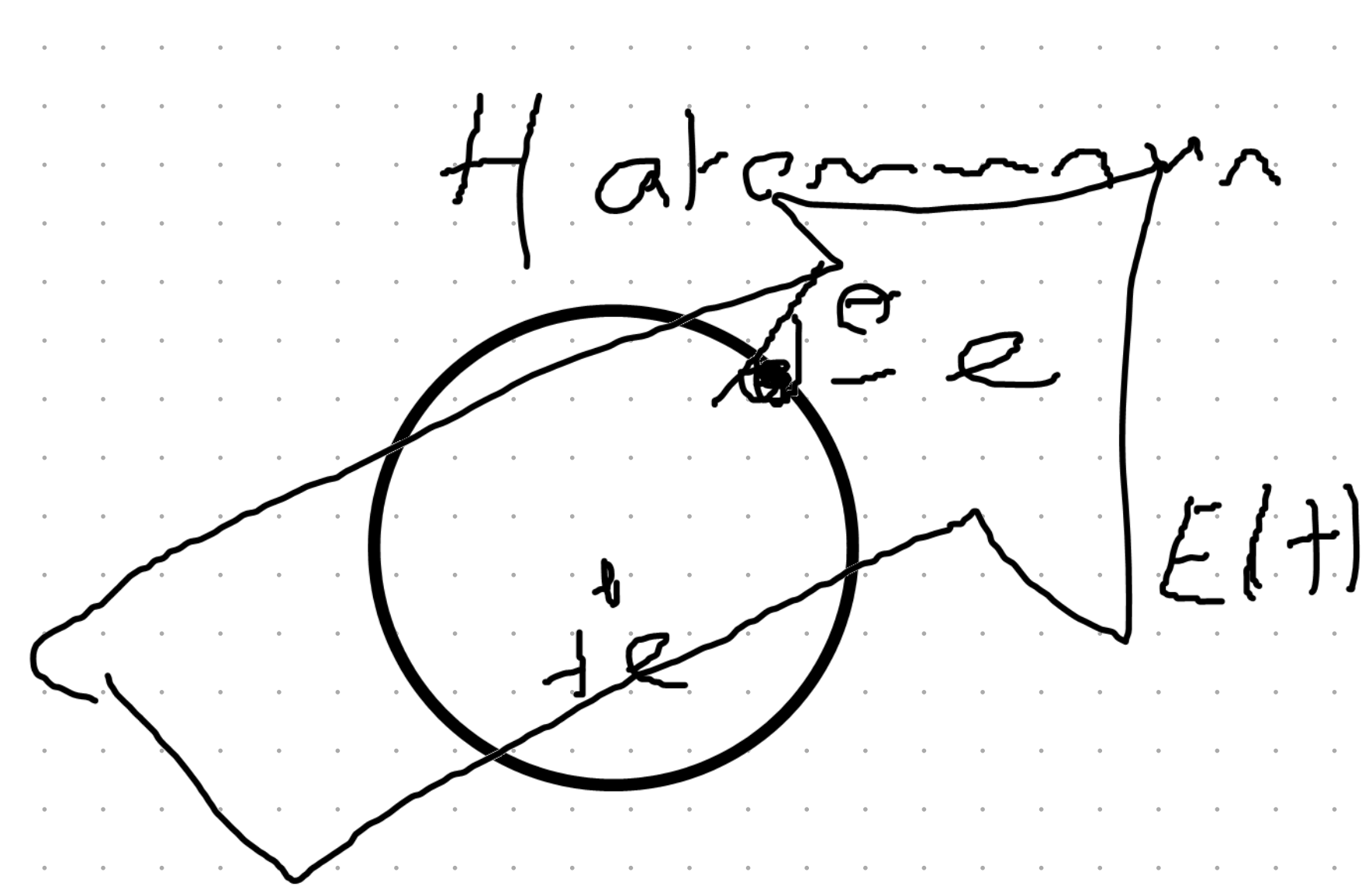
$$\lambda^1 \Rightarrow C_n^1(t) = \frac{\lambda}{i\hbar} \sum_n \underbrace{C_n^0(t)}_{\delta_{mn}} e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V | \phi_n \rangle \Rightarrow$$

$$\lambda^2 \Rightarrow C_n^2(t) = \frac{\lambda^2}{i\hbar} \sum_n C_n^1(t) e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V | \phi_n \rangle$$

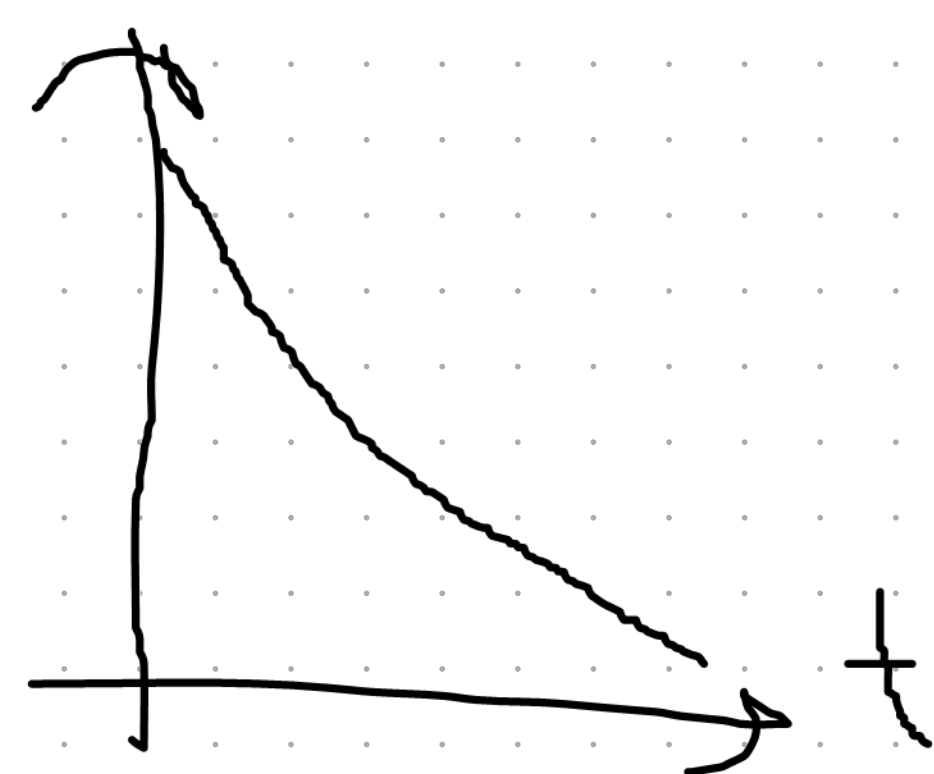
$$c_m'(t) = \frac{N}{i\hbar} \sum_n \frac{c_n^0}{\omega_{nm}} e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V | \phi_n \rangle \Rightarrow$$

$$c_m'(t) = \frac{N}{i\hbar} \int e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V | \phi_n \rangle dt$$

Soru T.D durumundaki H. Atomu $\vec{E}(t) = \vec{E}_0 e^{-t^2/\tau^2}$ elektrik alanının
müddetli bir etkisiyle, bir zaman sonra $(t) \gg \tau$ τ sist. karakteristiki-
gibi



H atomunun $n=2$ $l=1$ $m=0$ durumuna geçişleri



$$|c_p(t)|^2 \Rightarrow P = |1,1,m| = |2,1,0|$$

$$c_m'(t) = \frac{N}{i\hbar} \int e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \phi_m | V | \phi_n \rangle dt$$

$$\frac{N}{i\hbar} \int e^{-i(E_{210}^0 - E_{100}^0)t/\hbar} \langle \phi_{100} | V | \phi_{210} \rangle dt$$

$$\langle \phi_{100} | V | \phi_{210} \rangle = \int d^3r \phi_{100}^*(\vec{r}) V \phi_{210}(\vec{r})$$

$$= e |E_0| e^{-t^2/\tau^2} \cos \theta \Rightarrow$$

$$e |E_0| \int d^3r \phi_{100}^* z \phi_{210} \int e^{-t^2/\tau^2} e^{-i(E_{210}^0 - E_{100}^0)t/\hbar} dt$$

$$? \omega_{21} = \omega \quad \frac{1}{\tau^2} = b \Rightarrow$$

$$e |E_0| \int d^3r \phi_{100}^* z \phi_{210} \int e^{-bt^2} e^{-i\omega t/\hbar} dt$$

$$\Rightarrow -b \left(t^2 + \frac{\omega^2}{4b^2\hbar^2} \right) \quad \left(t + \frac{\omega}{2b\hbar} \right)^2 = t^2 + \frac{\omega t}{b\hbar} + \frac{\omega^2}{4b^2\hbar^2} \Rightarrow \left(t + \frac{\omega}{2b\hbar} \right)^2 - \frac{\omega^2}{4b^2\hbar^2} = t^2 + \frac{\omega t}{b\hbar}$$

$$-b \left\{ \left(t + \frac{\omega}{2b\hbar} \right)^2 - \frac{\omega^2}{4b^2\hbar^2} \right\}$$

$$-b \left\{ \left(t + \frac{d}{2b\hbar} \right)^2 - \frac{d^2}{4b^2\hbar^2} \right\}$$

$$e|E_0| \int dr^3 \phi_{100}^* z \phi_{210} \int e^{-\left(b t^2 + d t / \hbar \right)} dt$$

$$e|E_0| \int dr^3 \phi_{100}^* z \phi_{210} \int e^{-b \left\{ \left(t + \frac{d}{2b\hbar} \right)^2 - \frac{d^2}{4b^2\hbar^2} \right\}} dt$$

$$\int e^{-b \left(t + \frac{d}{2b\hbar} \right)^2 + \frac{d^2}{4b^2\hbar^2}} dt = \int e^{-b \left(t + \frac{d}{2b\hbar} \right)^2} \cdot e^{\frac{d^2}{4b^2\hbar^2}}$$

$$\Rightarrow e^{\frac{d^2}{4b^2\hbar^2}} \int e^{-b \left(t + \frac{d}{2b\hbar} \right)^2} dt$$

$$\sqrt{\frac{\pi}{b}} \Rightarrow 2\sqrt{\pi} e^{-\frac{d^2}{4b^2\hbar^2}} e|E_0| \langle \phi_{100} | z | \phi_{210} \rangle$$